Physics - Graduate Level Assessment 11 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A principal source of the Earth's internal energy is

    (a) electromagnetic radiation.
    (b) thermal conduction.
    (c) gravitational pressure.
    (d) radioactivity.
    (e) nuclear fusion.

2. The negative muon, mu$^{-}, haspropertiesmostsimilartowhichofthefollowing$?

    3. Gluon
    4. Electron
    5. Quark
    6. Proton
    7. Photon

How many nuclei are there in 1 kg of aluminum?

(a) 7.21 x 10^{2}5 nuclei
(b) 9.09 x 10$^{-}26nuclei$
(b) 4.48 x 10$^{-}26nuclei$
(b) 26.98 x 10$^{-}27nuclei$
(b) 3.01 x 10^{2}2 nuclei

The force required to start a mass of 50 kilograms moving over a rough surface is 343 Nt. What is the coefficient of starting friction?

(a) 0.75
(b) 0.40
(c) 0.55
(d) 0.65
(e) 0.50

How are planetary rings made?

(a) From accretion within the solar nebula at the same time the planets formed
(b) From fragments of planets ejected by impacts
(c) From dust grains that escape from passing comets
(d) From the remnants of a planet's atmosphere
(e) From the gravitational pull of nearby planets

True / False (10 marks)

Answer True or False

6. True or False: The centripetal force necessary for a 3200 pound car to travel around an 880-ft radius curve at 60 mph is 960 lb.
7. True or False: A minimum voltage of 65000 volts is sufficient to excite K radiation from a tungsten target, producing a K_alpha wavelength of 0.22 A.
8. True or False: The angular acceleration of the turntable is 0.58 sec$^{-}2, indicatingaslowerincreaseinspe$

8. True or False: Increasing the absolute temperature of a blackbody by a factor of 3 results in a decrease in the energy radiated per second per unit area.
9. True or False: The magnetic field lines around a current-carrying wire form concentric circles that radiate outward from the wire.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between voltage and energy dissipation in a resistor. If the voltage across a resistor that dissipates energy at 1 W is doubled, what is the new rate of energy dissipation? Explain your reasoning.
12. Calculate the time period of oscillation and the angular frequency of an ultrasonic transducer operating at 6.7 MHz, and discuss the significance of these values in medical diagnostics.

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